National Institute of Technology Warangal
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EC418 Biomedical Instrumentation and Signal Processing
Mini Research Project (Assignment 2): Medical Signal Analysis and Disease Diagnosis

Physiology-Aligned Multiresolution Analysis for Patient-Independent Heart-Murmur Detection:

A Comparative Study of Wavelet and Fourier Features

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Code, data provenance and every saved result: `https://github.com/pm2m1/ec418-circor-research`

Abstract

We study patient-independent detection of heart murmurs from phonocardiograms (PCG) on the public PhysioNet CirCor DigiScope dataset, comparing wavelet-packet (WPD), short-time Fourier (STFT) and wavelet-scattering features under cardiac-cycle-aligned and whole-signal designs, with WEKA classifiers on frozen subject-grouped folds (874 patients) and hypotheses registered before each analysis. Three findings stand out. (i) Cardiac-cycle timing carries the information: with the same features, destroying the alignment to the expert S1/systole/S2/diastole labels lowers patient AUROC by +0.131 to +0.174 (all 25 folds). (ii) The transform does not matter: wavelet packets were never better than matched STFT bands (differences −0.042 to +0.003), equivalence was not shown, and wavelet scattering did not beat STFT bands. (iii) A whole-signal STFT model trained on CirCor reaches AUROC 0.709 [0.675, 0.741] for congenital heart disease (CHD) on the independent ZCHSound cohort, with the physiological subtype order. An automatic pipeline with a learned span detector comes within the pre-registered −0.03 of expert labels on the point estimate, but statistical non-inferiority is not established, and its advantage over a whole-signal baseline that is given the same span information is not established either (+0.046 [−0.001, +0.093]). Negative results, errata and post-hoc analyses are reported as such; nothing here supports clinical deployment.

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1 Introduction

Cardiac auscultation is the cheapest screen for structural heart disease, but it depends on the listener's skill. Digital stethoscopes record the PCG, and automatic analysis could support screening where specialists are scarce. Many PCG pipelines segment each recording into cardiac cycles and describe each cycle phase with time–frequency features; wavelet transforms are a common choice because murmurs are short, non-stationary sounds [4].

This project asks a narrow question that such pipelines usually leave open: when a cycle-aligned wavelet representation helps, is it because of the *wavelet* or because of the *alignment* with the cardiac cycle? A second question follows from the first: how much of a result depends on the expert segmentation that the CirCor release provides, and that a deployed system would not have? We answer both with pre-registered contrasts on frozen, subject-grouped folds, and we report every hypothesis, including the ones that failed.

2 Clinical problem

A murmur is an additional sound produced by turbulent blood flow, heard in systole or diastole between the first (S1) and second (S2) heart sounds. Murmurs can be innocent or caused by structural disease such as septal defects (VSD, ASD), a patent ductus arteriosus (PDA) or valve lesions. Murmur detection is therefore a **screening** task: a detected murmur triggers referral for echocardiography, which establishes the diagnosis. This report addresses murmur detection only. The CirCor "Outcome" variable (clinical outcome) and the ZCHSound CHD labels are different targets; we never treat them as murmur labels.

3 Literature review (2024–2026)

The George B. Moody PhysioNet Challenge 2022 made CirCor the reference dataset for murmur detection [1, 3]. Its winning entry combined a recurrent network with parallel hidden semi-Markov models that segment and classify at the same time [5]; HSMM segmentation of heart sounds follows Springer et al. [6]. Table 1 summarises recent work. The numbers are quoted as the authors report them and are **not on a common scale**: they differ in metric, label set (two or three classes), unit (snippet, recording or patient), split and in whether expert annotations are used on the test data.

4 Research gaps

1. Wavelet and Fourier features are rarely compared with *matched* band edges, time grid and dimension, so a reported wavelet advantage may come from a different feature design rather than from the transform.
2. Cycle-aligned results are often obtained with the expert segmentation on the test data (Table 1); how much of the result that annotation is worth is seldom measured.
3. Several pipelines split recordings or snippets rather than patients; CirCor has several recordings per child and linked visits, so such splits can leak identity.
4. Automatic segmentation can introduce shortcuts: which parts of a recording a segmenter keeps may itself correlate with the label.

Table 1: Recent CirCor murmur-detection work (2024–2026). Headline values as reported by the authors (source: abstract or the stated section; `references/g3b/case_study_numbers.csv`). "Expert seg.": whether expert annotations select or cut the test data.

Work	Method	Expert seg. at test	Evaluation	Headline (as reported)
Vimalajeewa et al. 2025 [4] (*Sci. Rep.*)	multiscale wavelet features (fractal scaling, entropy)	no	recording level, random split	re-evaluated here under our folds (Sec. 11)
McDonald et al. 2024 [5] (*PLOS DH*)	RNN + parallel HSMMs, joint segmentation	no	Challenge hidden test	Challenge winner
Niizumi et al. 2024 [7] (EMBC)	pre-trained general-purpose audio representations	no	patient level	W.acc 0.832
Dehghan Manshadi & Mihandoost 2024 [8] (*Sci. Rep.*)	Stockwell transform + deep features	not stated	patient level	accuracy 93 %
Nie et al. 2024 [9] (MLSP)	2-D CNN, multi-task with heart rate	yes (annotated periods only)	5-s snippets, subject-disjoint	accuracy 97.49 %
Hong et al. 2025 [10] (*IEEE JBHI*)	periodic-pattern-aware transformer, no cycle segmentation	no	10-fold CV (unit not checked)	W.acc 84.27 %
Kutlu & Çolak 2026 [11] (*J. Clin. Med.*)	eleven classifiers, multi-representation fusion	no	CirCor patient-disjoint CV; external zero-shot	external AUROC 0.738 (CinC 2016), 0.985 (Yaseen)
Ladislaus et al. 2026 [12] (arXiv)	HMS-Net and BiLSTM under noise; aggregation rules	no	5 patient-level folds, 3 classes	W.acc 0.8021
Nogueira et al. 2026 [13] (arXiv)	synchronous multi-channel CNN	yes (segments start at expert S1)	patient level, 10-fold	(Se+Sp)/2 = 0.965
Zewail 2026 [14] (LNCS)	training-free scattering transformer	not checked	not checked	W.acc 0.786

5 Dataset and annotation details

CirCor DigiScope v1.0.3 [1] (PhysioNet [2]): 942 patients aged up to 21 years, 3,163 recordings at 4 kHz from up to four auscultation sites (aortic, pulmonary, tricuspid, mitral), each patient labelled murmur Present, Absent or Unknown by an expert, with S1/systole/S2/diastole segmentation (`.tsv`) for annotated parts of each recording. We use Present vs Absent: **874 patients (179 Present, 695 Absent)**. Linked visits of the same child share an `Additional ID`; grouping by it gives 816 subjects, and folds never split a subject. The study uses the official PhysioNet download, verified file by file against PhysioNet's checksum list (10,434 files); Phase 1 used a byte-identical redistribution, and the recordings and murmur labels are identical in both. **ZCHSound** [16] (external test only, CC BY 4.0): 1,259 children, one recording each, labelled by echocardiography-confirmed diagnosis; the clean set has 941 children (408 with CHD), the noisy neonatal set 318.

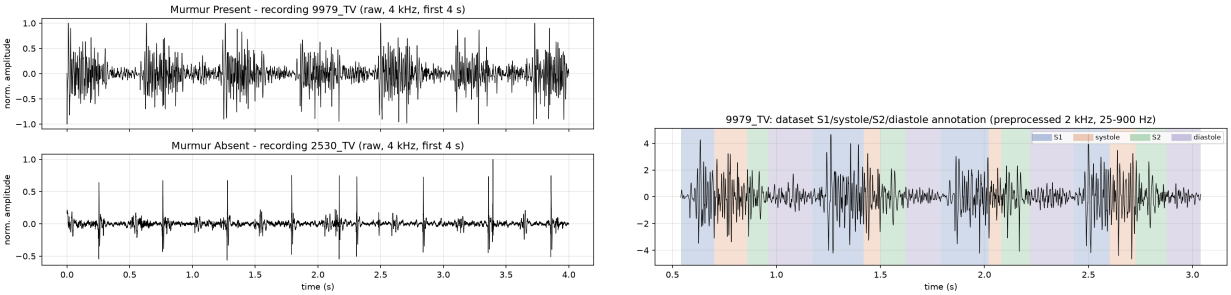


Figure 1: Left: raw PCG of a murmur-present and a murmur-absent recording. Right: expert S1/systole/S2/diastole annotation of the cardiac cycles (CirCor `.tsv`). Phase-1 figures.

6 Signal-processing methodology

Every recording is resampled to 2 kHz (polyphase FIR), band-pass filtered at 25–900 Hz (zero-phase Butterworth, order 4) and z-scored; no statistic is estimated across recordings. Figure 2 shows the pipeline.

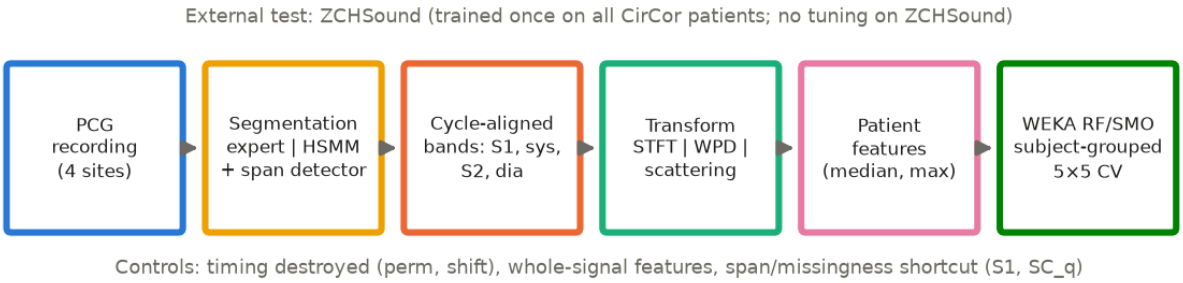


Figure 2: Pipeline and controls.

6.1 Wavelet transform mathematics

A signal x is represented in an orthonormal wavelet basis by scaling functions $\phi_{J,k}$ and wavelets $\psi_{j,k}$,

$$x(t) = \sum_k c_{J,k}\,\phi_{J,k}(t) + \sum_{j=1}^{J}\sum_k d_{j,k}\,\psi_{j,k}(t), \qquad \psi_{j,k}(t) = 2^{-j/2}\psi(2^{-j}t - k). \tag{1}$$

The discrete wavelet transform (DWT) computes the coefficients with a two-channel filter bank: with low-pass filter h and high-pass filter $g[n] = (-1)^n h[L-1-n]$,

$$a_{j+1}[k] = \sum_n h[n-2k]\,a_j[n], \qquad d_{j+1}[k] = \sum_n g[n-2k]\,a_j[n], \tag{2}$$

starting from $a_0 = x$. Each level halves the band and the sampling rate, so the DWT refines only the low-frequency branch (Figure 3). We use the Daubechies db4 filter (length 8), fixed before any result for its time localisation over the 100–300 ms systolic interval; db6, sym8 and coif3 were sensitivity analyses only.

6.2 Wavelet-packet decomposition

The wavelet packet decomposition (WPD) splits *both* the approximation and the detail branch at every level. At depth J it yields 2^J nodes; ordered by frequency, node b covers approximately

$[\,b\,f_s/2^{J+1},\,(b+1)\,f_s/2^{J+1})$. With $f_s = 2\,\text{kHz}$ and $J = 4$ this gives 16 bands of 62.5 Hz between 0 and 1 kHz on an 8 ms grid (periodisation mode). Band energy maps $E_b(t)$ are the squared node coefficients. Real filters leak between neighbouring bands (Figure 4); the STFT comparison below uses the same nominal band edges, so leakage is part of what is compared.

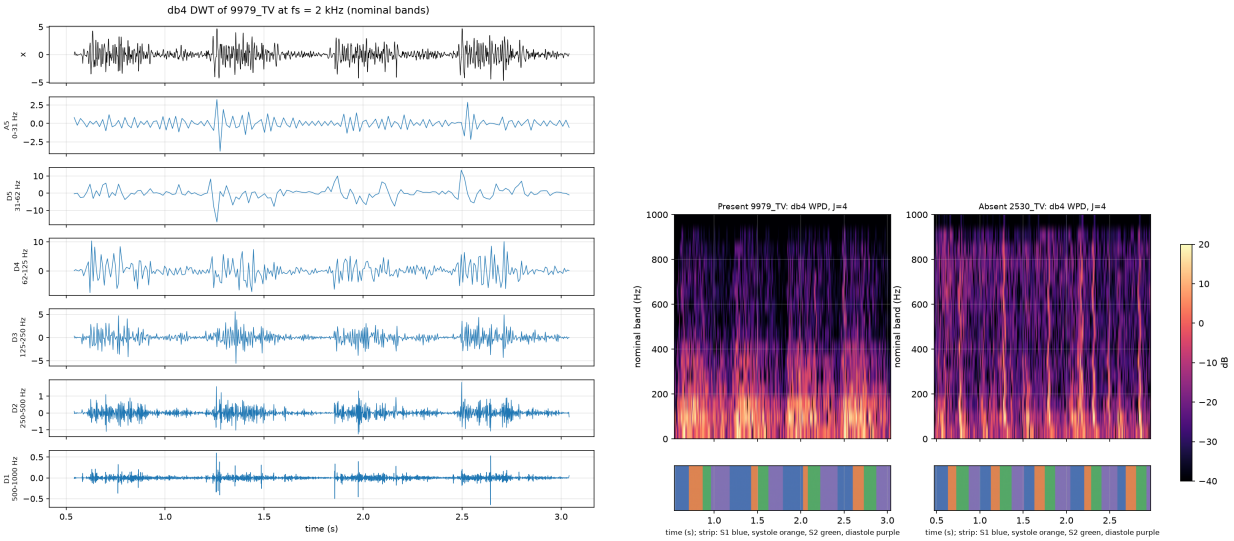


Figure 3: Left: DWT decomposition of a PCG. Right: depth-4 WPD time–frequency energy map. Phase-1 figures.

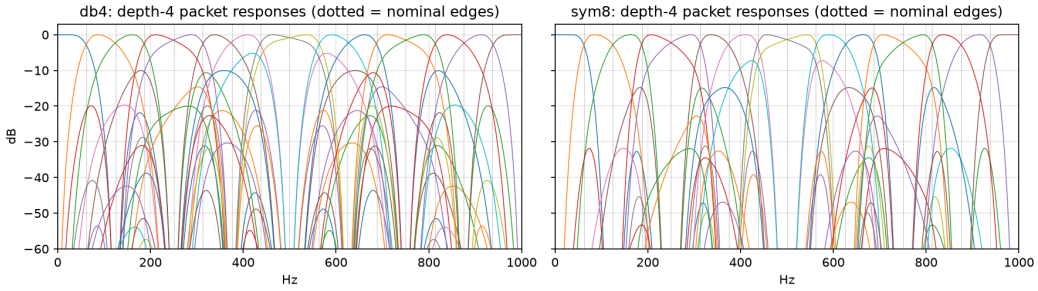


Figure 4: Frequency responses of the 16 depth-4 packets for db4 and sym8 (dotted: nominal edges).

6.3 STFT comparison

The STFT $X(m,\omega) = \sum_n x[n]\,w[n-mR]\,e^{-j\omega n}$ (Hann window w, hop R) is pooled into the *same* 16 frequency bands on the *same* time grid, giving band energy maps of identical dimension. Hence "WPD vs STFT" changes only the transform. *wpds* is WPD with the STFT's Hann window applied as temporal smoothing, which tests whether a difference is only estimator variance (post-hoc in Phase 1, pre-registered in Phase 2). Wavelet scattering [17] (kymatio [18], $J = 6$, $Q = (8, 1)$, first and first+second order) was added in Phase 3 under the same band features.

6.4 Features

Whole-signal (W): per band, log relative band energy and $\log(q_{90}/q_{50})$ burstiness, plus two entropies. *Cycle-aligned* (A, E2, E4): per band, log(systolic energy / S1 energy) and log(systolic / diastolic energy), plus two entropies. A recording contributes aligned features only if enough systole, diastole and S1 are annotated; otherwise they are missing (recorded, not imputed silently). Patient vector: median and maximum over the patient's recordings. Baselines: B0 (MFCC and spectral descriptors) and B1, the 2025 wavelet baseline [4], re-implemented from the authors' code.

7 Cardiac-cycle segmentation

- **Expert labels** (E2 `true`): the CirCor `.tsv` annotation. Results using it at test time are *oracle-conditioned*. Controls keep the features and destroy timing: `perm` permutes the labels within a recording, `shift` shifts them circularly.
- **E1**: homomorphic-envelope peak picking, constants fixed before the run.
- **E1b**: a Python port of the logistic-regression HSMM of Springer et al. [6], trained on the training folds and applied to whole recordings.
- **E4**: the LR-HSMM plus a learned span detector (logistic regression on six features of 1-s windows) that keeps windows likely to be annotatable; trained on the training folds only.
- **E4q** (Phase 3): E4 with the same fraction of windows kept in every recording, which removes the span-fraction shortcut by construction.
- **Diagnostics D1–D3** use the expert spans at test time (labelled as such): HSMM timing inside expert spans (D1), timing destroyed inside spans (D2), whole-signal features inside spans (D3).

8 WEKA classification

All classifiers are WEKA 3.9.7 [19], called in batch from Python: RandomForest [20] (`-I 300 -K 0 -S 1 -num-slots 1`; primary), SMO [21] with a linear PolyKernel and logistic calibration (secondary), and MLP and Logistic in Phase 1. WEKA's own cross-validation is never used, because it splits rows and would put recordings of the same child in training and test data. One fold reproduced with the WEKA command line equals the pipeline's predictions exactly (699 training and 175 test patients; `weka_demo/`).

9 Experimental protocol

Folds: 5 repeats × 5 outer folds, subject-grouped and stratified, fixed before any model ran (sha256 `c4aa74c7...2339`). Thresholds are chosen on inner training folds only (specificity ≥ 0.80). Phase 1 (v1) was locked before any result; Phase 2 (E0–E4, D1–D3, S1) and Phase 3 (G1–G4) were each pre-registered with fixed decision rules before their runs, but after the preceding results on the same folds (sequential design). Deviations are dated amendments committed before the affected run.

10 Evaluation metrics

Patient AUROC, the probability that a random Present patient scores higher than a random Absent patient. We report two definitions and never mix them: the *mean of repeat AUROCs* (± SD over the 5 repeats), and the *AUROC of repeat-averaged scores*, where each patient's out-of-fold score is averaged over the repeats; the latter is used by every hypothesis test, with a 95 % subject-bootstrap CI (2000 resamples). At the operating point we report sensitivity (Se), specificity (Sp) and the Challenge weighted accuracy, which in the binary encoding is

$$\text{W.acc} = \frac{5\,\text{TP} + \text{TN}}{5\,(\text{TP} + \text{FN}) + (\text{TN} + \text{FP})}. \tag{3}$$

Decision rules (registered per hypothesis): typically $\Delta \geq 0.03$, CI lower bound > 0 and at least 18/25 folds with $\Delta > 0$. H6a, H8a and H9a used a point-estimate criterion $\Delta \geq -0.03$; it is *not* a formal non-inferiority test.

11 Results

Table 2 summarises the reference models. Cycle-aligned WPD beats whole-signal WPD (H3: +0.113 [+0.065, +0.165]) and the re-evaluated 2025 baseline B1 (H4b: +0.072 [+0.027, +0.122]), but not B0 (H4a: +0.051 [−0.002, +0.102]). Figure 5 shows the patient ROC curves.

Table 2: Main models (RF, patient level, AUROC of repeat-averaged scores; operating point: inner-fold threshold for specificity ≥ 0.80; W.acc at that threshold).

Model (RF)	AUROC [95 % CI]	Se	Sp	W.acc
B0 acoustic baseline	0.747 [0.699, 0.793]	0.573	0.815	0.679
B1 Vimalajeewa et al.	0.726 [0.677, 0.773]	0.551	0.813	0.666
whole-signal STFT (W_stft)	0.751 [0.701, 0.794]	0.577	0.808	0.678
expert-aligned STFT (oracle)	0.815 [0.772, 0.853]	0.663	0.810	0.727
E4 automatic STFT	0.809 [0.766, 0.849]	0.679	0.798	0.731
E4q shortcut-free STFT	0.799 [0.756, 0.842]	0.637	0.810	0.713

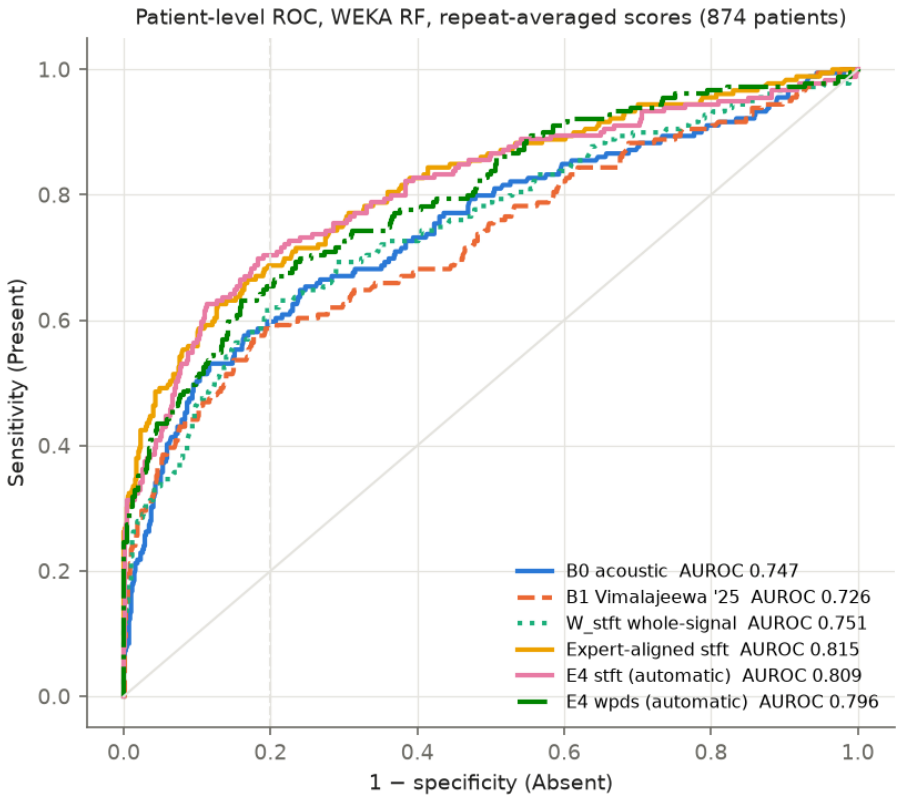


Figure 5: Patient-level ROC (RF, repeat-averaged scores).

11.1 Ablation: timing versus span choice

With expert labels, destroying timing costs +0.131 to +0.174 AUROC (`perm`, H5a–c) and +0.192 to +0.206 (`shift`, H5d), all in 25/25 folds. The diagnostics split this exactly (Figure 6): within-span timing is worth +0.169 to +0.193, and span selection without timing contributes −0.062 to −0.020. This decomposition is against the timing-destroyed null; it does not mean that alignment explains every difference between models.

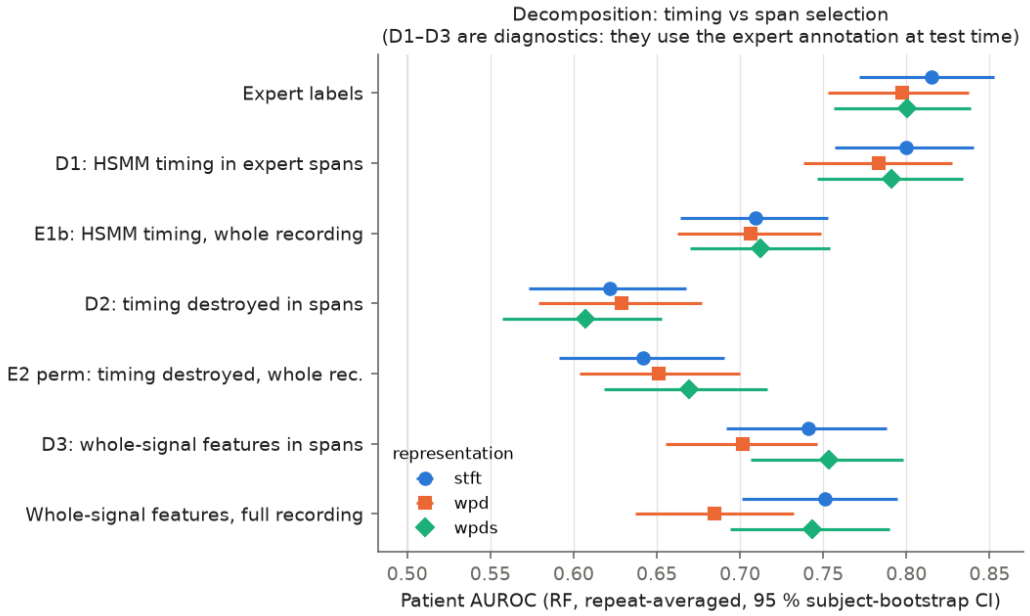


Figure 6: Decomposition of the expert-label result (RF AUROC, 95 % CI). D1–D3 use the expert annotation at test time.

11.2 Wavelet versus Fourier

Under every labelling, WPD and smoothed WPD are at most equal to matched STFT bands (Figure 7; H2a: −0.017 [−0.034, 0.000]; H2b, whole signal: −0.066 [−0.095, −0.036]); equivalence within ±0.03 is not shown (−0.015 [−0.032, +0.003]). Wavelet scattering does not beat STFT bands either (Figure 7; H11a −0.026 [−0.058, +0.005], H11b −0.013 [−0.044, +0.017]); second-order paths add +0.013 [+0.001, +0.025], below the registered 0.03.

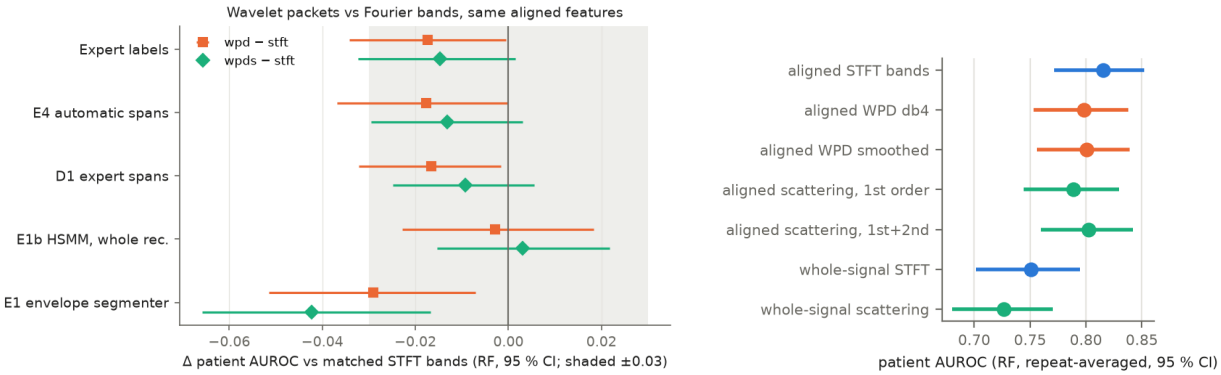


Figure 7: Left: WPD minus matched STFT bands under each labelling (shaded ±0.03). Right: STFT, WPD and scattering features (RF AUROC).

11.3 Failed hypotheses

Reported in full in Appendix A. Not supported: H1 (no material inflation from random splits in the authors' recording-level setting), H2a/H2b (no wavelet advantage), H4a (no gain over B0), every E1 and E1b hypothesis (H6, H8), H9c for the primary wpd, H11a–c (scattering), and H12a (shortcut-free advantage). The planned soft-murmur experiment (H7) was dropped by amendment before it ran.

11.4 Automatic-segmentation analysis

Envelope peaks (E1) and the LR-HSMM on whole recordings (E1b) miss their targets (e.g. H8a −0.091 [−0.135, −0.045]). The HSMM's timing is close to expert timing inside expert spans (expert − D1: +0.010 to +0.015); what is missing is span selection (D1 − E1b: +0.077 to +0.091). With the learned span detector (E4) the point estimate is within the pre-registered −0.03 of expert labels (H9a: −0.007 [−0.044, +0.031] for wpd), but the CI lower bound is below −0.03, so statistical non-inferiority is **not established**. E4 beats B1 (H9b: +0.065 to +0.083). Table 3 gives the cost of replacing the expert segmentation by each automatic one (conditioning gaps, [15]).

Table 3: Conditioning gaps: AUROC(expert) minus AUROC(estimated or random) (RF, 95 % CI).

Gap (RF AUROC)	stft	wpd	wpds
Δ_{or} expert − perm	0.174 [0.115, 0.229]	0.147 [0.094, 0.199]	0.131 [0.079, 0.185]
Δ_{or} expert − shift	0.202 [0.152, 0.254]	0.192 [0.141, 0.248]	0.206 [0.153, 0.260]
Δ_{oe} expert − E1	0.050 [0.011, 0.090]	0.062 [0.021, 0.102]	0.077 [0.037, 0.120]
Δ_{oe} expert − E1b	0.106 [0.063, 0.148]	0.091 [0.045, 0.135]	0.088 [0.045, 0.130]
Δ_{oe} expert − E4	0.006 [−0.031, 0.045]	0.007 [−0.032, 0.044]	0.005 [−0.035, 0.044]
Δ_{oe} expert − E4q	0.016 [−0.023, 0.055]	0.006 [−0.034, 0.047]	0.012 [−0.029, 0.053]

11.5 Shortcut-learning investigation

E4's selected-span fraction alone predicts the class (AUROC 0.646, post-hoc check). Against plain whole-signal STFT, the H9c rule is met for stft (+0.058 [+0.011, +0.108]) but not for the primary wpd (+0.040 [−0.003, +0.089]). When whole-signal STFT is given the same span and missingness variables (S1), E4's margin is +0.026 to +0.043 with CIs that include zero. The shortcut-free E4q removes the shortcut (shortcut variables alone: 0.482 [0.434, 0.529]; test recordings without aligned features fall from 15.8 % to 0.6 % for Absent patients) at essentially no cost (E4q − E4: −0.009 [−0.041, +0.019]), and learned selection beats random windows (H12b: +0.066 [+0.030, +0.100]). Its margin over whole-signal STFT with the same information is +0.046 [−0.001, +0.093] (H12a): the CI includes zero, so this advantage is **not established**.

11.6 External test on ZCHSound

A whole-signal STFT model trained once on all 874 CirCor patients was applied unchanged to ZCHSound (Figure 8, Table 4). On the clean set the any-CHD AUROC is 0.709 [0.675, 0.741] (H10a supported; internal CirCor 0.751). By subtype: PDA 0.854 [0.796, 0.905], VSD 0.785 [0.744, 0.823], ASD 0.659 [0.605, 0.716], PFO 0.522 [0.447, 0.596], the pre-registered physiological order (H10b: VSD − PFO +0.263 [+0.180, +0.343]). On the noisy neonatal set the AUROC is 0.535 [0.471, 0.597] (chance). A post-hoc check shows that the score is unrelated to age (Spearman −0.028).

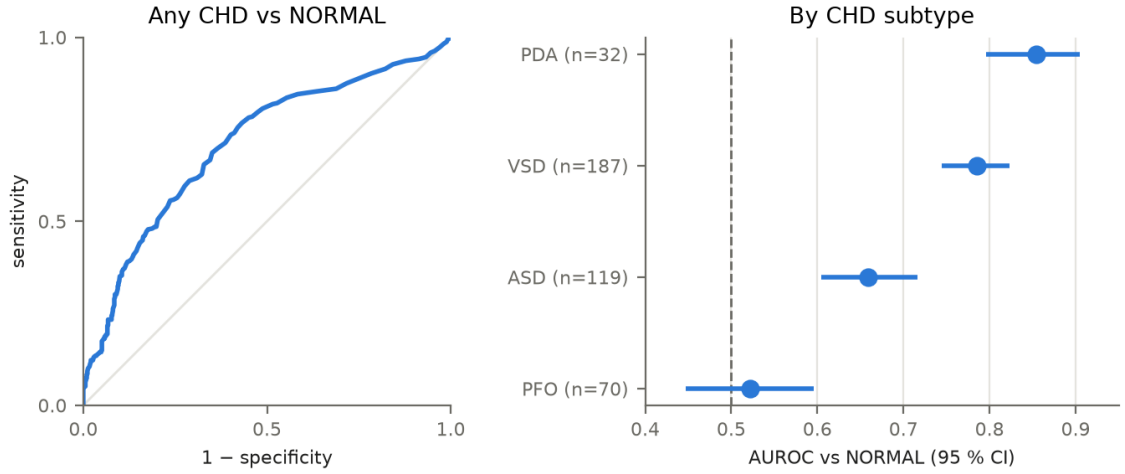


Figure 8: ZCHSound clean set, RF W_stft trained on CirCor: any-CHD ROC and AUROC by subtype vs NORMAL (95 % CI).

Table 4: External AUROC for any CHD (RF). B0 enters no verdict: its external signal is an artefact.

Model (RF, trained on CirCor)	clean	noisy	CirCor internal
W_stft (pre-registered, H10a)	0.709 [0.675, 0.741]	0.535 [0.471, 0.597]	0.751
W_wpd_db4_smooth	0.784 [0.751, 0.813]	0.497 [0.436, 0.562]	0.743
W_scat1	0.559 [0.521, 0.597]	0.411 [0.347, 0.471]	0.727
B0 (artefact, no verdict)	0.699 [0.662, 0.732]	0.503 [0.438, 0.570]	0.747

11.7 Comparison with recent work

Only B1 [4] was re-evaluated on our data and folds; on that common scale our aligned and automatic models are better (H4b, H9b). The higher numbers in Table 1 are not comparable with ours. For example, Nogueira et al. [13] report $(\text{Se}+\text{Sp})/2 = 0.965$ with test segments that start at expert-annotated S1 onsets; on the same balanced-accuracy scale our expert-label model reaches 0.736, E4 0.739. Nie et al. [9] report snippet-level accuracy on annotated periods and note themselves that this differs from recording-level Challenge evaluation. In our data the expert segmentation is worth up to 0.106 AUROC over a naive automatic pipeline, which illustrates how much test-time conditioning can change a number.

12 Critical discussion

The wavelet experiments are informative even though STFT was at least as good: with matched bands, grid and features, the transform was the only change. For these features, therefore, the transform itself brings no advantage; a wavelet advantage reported in other pipelines would need another explanation (for example a different feature design or test-time conditioning), which we did not test. The strongest effect is timing, but it was measured with expert labels; making it work automatically needed span selection, and part of that apparent gain is explained by which spans are kept. The honest summary of the automatic pipeline is therefore “close to expert labels on the point estimate, not shown to be better than a shortcut-aware whole-signal baseline”. The external test supports transfer of a simple whole-signal model across cohorts and devices, with physiologically sensible behaviour (loud shunts detected, silent PFO not), but it tests a CHD label, not murmur detection.

13 Limitations

- Cross-validation on one public cohort; no CirCor hidden test set; no prospective or clinical evaluation.
- No intrinsic wavelet superiority was demonstrated; STFT bands were at least as good.
- Aligned results depend on segmentation and span selection; expert-label results are oracle-conditioned.
- Only 179 Present patients; CIs are wide.
- Possible shortcut learning in automatic pipelines (S1; H12a not established).
- The Unknown class is excluded, so the task is easier than the three-class Challenge task.
- ZCHSound labels are CHD diagnoses, not murmurs; the noisy neonatal subset is at chance.
- SMO is numerically unstable on scattering features; no SMO scattering number is used as evidence.
- E1b, E4, S1 and Phase 3 were designed after earlier results on the same folds.
- Murmur detection is not a clinical-outcome diagnosis; nothing here supports deployment.
- The systematic literature audit was searched and screened but not coded, so no literature-wide rate is reported.

14 Conclusion

On CirCor, cardiac-cycle timing carries the murmur information available to these features, and the choice between wavelet packets, STFT and scattering does not change the result. A whole-signal model transfers to an independent paediatric CHD cohort with the expected physiological ordering. Future work: signal-quality gating for noisy recordings, learned models evaluated under the same subject-grouped and shortcut-controlled protocol, a murmur-labelled external set, and a systematic audit of test-time expert segmentation in published CirCor work.

Reproducibility statement

Code, protocols, saved predictions and tables: `https://github.com/pm2m1/ec418-circor-research`. Protocol tags `v1.0-pilot`, `protocol-v2-lock`, `e1b-prereg`, `diag-prereg`, `e4-prereg`, `s1-prereg`; results `results-final` and `phase3-results`; checksum manifests in `results/v1_freeze/` and `results/final_freeze/`; quick verification with `scripts/p5_verify_integrity.py`; full guide `docs/REPRODUCIBILITY.md`. Every number in this report is generated from saved tables (`scripts/p4_numbers.py`) and checked by `scripts/p4_check_numbers.py`.

AI-assistance statement

This project was carried out with substantial assistance from an AI coding assistant (Anthropic's Claude, used through Claude Code). It wrote most of the code, ran the analyses, drafted this report and the slides, and performed the literature screening and the transcription of the Google Scholar export. The author defined the study, made the decisions recorded in the protocols and amendments, and is responsible for the content. [The author should review this report and restate the findings in their own words before submission.]

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A Every pre-registered hypothesis

Table 5: All pre-registered hypotheses (RF, AUROC of repeat-averaged patient scores, subject-bootstrap 95 % CI). "SUPPORTED" for H6a/H8a/H9a means that the point-estimate rule was met.

Hyp.	Contrast (RF)	Δ AUROC [95 % CI]	folds > 0	verdict
H1	B1+SVM recording AUC: random split - subject-grouped	+0.015 [−0.004, +0.035]	n/a	NOT SUPPORTED
H2a	A_wpd_db4 - A_stft	−0.017 [−0.034, 0.000]	3/25	NOT SUPPORTED
H2b	W_wpd_db4 - W_stft	−0.066 [−0.095, −0.036]	1/25	NOT SUPPORTED
H3	A_wpd_db4 - W_wpd_db4	+0.113 [+0.065, +0.165]	25/25	SUPPORTED
H4a	A_wpd_db4 - B0	+0.051 [−0.002, +0.102]	21/25	NOT SUPPORTED
H4b	A_wpd_db4 - B1	+0.072 [+0.027, +0.122]	24/25	SUPPORTED
H5a	E2_wpd_true - E2_wpd_perm	+0.147 [+0.094, +0.199]	25/25	SUPPORTED
H5b	E2_stft_true - E2_stft_perm	+0.174 [+0.115, +0.229]	25/25	SUPPORTED
H5c	E2_wpds_true - E2_wpds_perm	+0.131 [+0.079, +0.185]	25/25	SUPPORTED
H5d	E2_wpd_true - E2_wpd_shift	+0.192 [+0.141, +0.248]	25/25	SUPPORTED
H5d	E2_stft_true - E2_stft_shift	+0.202 [+0.152, +0.254]	25/25	SUPPORTED
H5d	E2_wpds_true - E2_wpds_shift	+0.206 [+0.153, +0.260]	25/25	SUPPORTED
H6a	E1_wpd_auto - E2_wpd_true	−0.062 [−0.102, −0.021]	3/25	NOT SUPPORTED
H6b	E1_wpd_auto - B1	+0.010 [−0.038, +0.060]	15/25	NOT SUPPORTED
H6c(a)	E1_stft_auto - E2_stft_true	−0.050 [−0.090, −0.011]	1/25	NOT SUPPORTED
H6c(b)	E1_stft_auto - B1	+0.039 [−0.009, +0.088]	20/25	NOT SUPPORTED
H6c(a)	E1_wpds_auto - E2_wpds_true	−0.077 [−0.120, −0.037]	1/25	NOT SUPPORTED
H6c(b)	E1_wpds_auto - B1	−0.003 [−0.051, +0.047]	12/25	NOT SUPPORTED
H8a	E1b_wpd_auto - E2_wpd_true	−0.091 [−0.135, −0.045]	1/25	NOT SUPPORTED
H8b	E1b_wpd_auto - B1	−0.019 [−0.064, +0.029]	6/25	NOT SUPPORTED
H8c(a)	E1b_stft_auto - E2_stft_true	−0.106 [−0.148, −0.063]	0/25	NOT SUPPORTED
H8c(b)	E1b_stft_auto - B1	−0.017 [−0.061, +0.031]	11/25	NOT SUPPORTED
H8c(a)	E1b_wpds_auto - E2_wpds_true	−0.088 [−0.130, −0.045]	0/25	NOT SUPPORTED
H8c(b)	E1b_wpds_auto - B1	−0.013 [−0.057, +0.034]	10/25	NOT SUPPORTED
H9a	E4_wpd - E2_wpd_true	−0.007 [−0.044, +0.031]	10/25	SUPPORTED
H9b	E4_wpd - B1	+0.065 [+0.022, +0.115]	25/25	SUPPORTED
H9c	E4_wpd - W_stft	+0.040 [−0.003, +0.089]	21/25	NOT SUPPORTED
H9a	E4_stft - E2_stft_true	−0.006 [−0.044, +0.031]	8/25	SUPPORTED
H9b	E4_stft - B1	+0.083 [+0.036, +0.135]	24/25	SUPPORTED
H9c	E4_stft - W_stft	+0.058 [+0.011, +0.108]	22/25	SUPPORTED
H9a	E4_wpds - E2_wpds_true	−0.005 [−0.044, +0.035]	9/25	SUPPORTED
H9b	E4_wpds - B1	+0.070 [+0.027, +0.120]	25/25	SUPPORTED
H9c	E4_wpds - W_stft	+0.045 [+0.001, +0.094]	21/25	SUPPORTED
equivalence	E2_wpds_true - E2_stft_true	−0.015 [−0.032, +0.003]	5/25	equivalence NOT shown
H7	soft murmurs (grade I/VI sensitivity)	–	-	NOT RUN (dropped, amendment E3-drop)
S1(a)	E4_stft - S1_Wstft_SC	+0.043 [−0.005, +0.096]	21/25	possibly shortcut-driven
S1(a)	E4_stft - S1_Wwpd_SC	+0.095 [+0.043, +0.149]	23/25	not explained by the span/missingness shortcut
S1(a)	E4_wpd - S1_Wstft_SC	+0.026 [−0.020, +0.075]	21/25	possibly shortcut-driven
S1(a)	E4_wpd - S1_Wwpd_SC	+0.077 [+0.027, +0.131]	24/25	not explained by the span/missingness shortcut
S1(a)	E4_wpds - S1_Wstft_SC	+0.030 [−0.015, +0.081]	18/25	possibly shortcut-driven
S1(a)	E4_wpds - S1_Wwpd_SC	+0.082 [+0.031, +0.137]	23/25	not explained by the span/missingness shortcut
H12a	E4q − (W_stft + SC_q), stft	+0.046 [−0.001, +0.093]	23/25	possibly shortcut-driven
H12a	E4q − (W_stft + SC_q), wpd	+0.038 [−0.010, +0.086]	21/25	possibly shortcut-driven
H12a	E4q − (W_stft + SC_q), wpds	+0.035 [−0.013, +0.083]	20/25	possibly shortcut-driven
H12b	E4q − rand-q, stft	+0.066 [+0.030, +0.100]	23/25	learned selection matters
H12b	E4q − rand-q, wpd	+0.071 [+0.037, +0.102]	22/25	learned selection matters
H12b	E4q − rand-q, wpds	+0.066 [+0.031, +0.099]	22/25	learned selection matters
H12c (desc.)	E4q − snr-q, stft	+0.020 [−0.006, +0.046]	18/25	descriptive
H12c (desc.)	E4q − snr-q, wpd	+0.013 [−0.013, +0.040]	17/25	descriptive
H12c (desc.)	E4q − snr-q, wpds	+0.007 [−0.021, +0.033]	10/25	descriptive
H11a	A_scat1 − A_stft	−0.026 [−0.058, +0.005]	4/25	NOT SUPPORTED
H11b	A_scat12 − A_stft	−0.013 [−0.044, +0.017]	7/25	NOT SUPPORTED
H11c	A_scat12 − A_scat1	+0.013 [+0.001, +0.025]	18/25	NOT SUPPORTED
H10a	ZCHSound clean any-CHD AUROC, W_stft	0.709 [0.675, 0.741]	n/a	SUPPORTED
H10b	AUROC(VSD) − AUROC(PFO)	+0.263 [+0.180, +0.343]	n/a	SUPPORTED
H10b	AUROC(VSD) − AUROC(ASD)	+0.126 [+0.064, +0.184]	n/a	SUPPORTED